电路样卷2 详解：

一、

1. 略.

2. 齐性定理法：假设2kΩ上流出1A电流，我们逐步倒推$U_S(t)$.

$$\Rightarrow \quad U_S(t) = 54V.$$

$$\therefore \quad \frac{27e^{-t}\sin 2t}{54} = \frac{I}{1} \quad \therefore I = \frac{1}{2}e^{-t}\sin t$$

$$\therefore U_{2k\Omega} = e^{-t}\sin 2t \ V.$$

（若电阻数一般，此方法优势明显）.

3. 假设出一个电流I： 标出个别未知元件电压

KVL： $10 - 10I - 4 - 30I + 20I = 0$

$$\therefore I = \frac{3}{10}A.$$

$$\therefore P = \frac{3}{10} \times 10 = 3W.$$

4. 对R两端戴维南：

U_{oc}： $24 \times \frac{3}{6+3} = 8V.$

R_{eq}： 置零24V电压源后，4Ω被短路. $R_{eq} = \frac{6\times 3}{6+3} = 2\Omega.$

$$\therefore P_{max} = \frac{U_{oc}^2}{4R_{eq}} = 8W, \text{ 取等: } R = R_{eq} = 2\Omega.$$

5. 节点2： $(3+4)U_{n2} - 3U_{n1} = 9V$

补充方程： $V = U_{n2} - U_{n1}.$

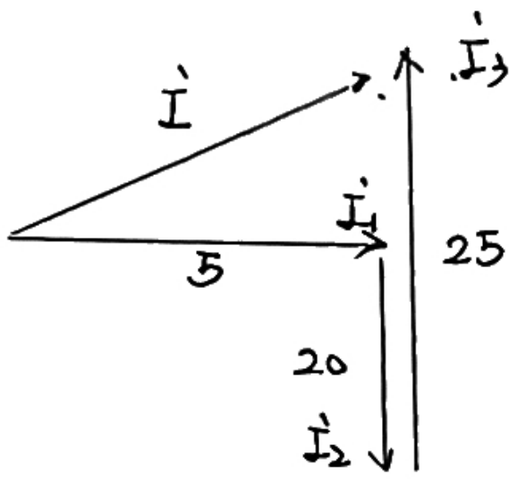
$$\Rightarrow \quad 7U_{n2} - 3U_{n1} = 9(U_{n2} - U_{n1})$$

$$\Rightarrow \quad (7-9)U_{n2} = (3-9)U_{n1}.$$

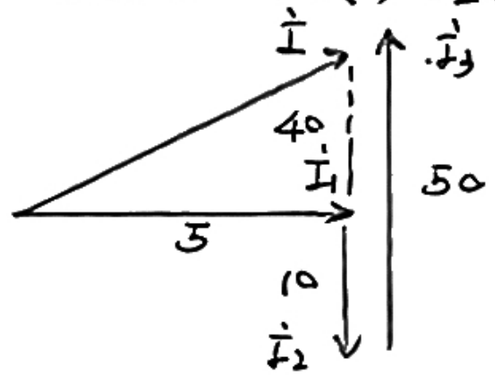
$$\therefore \frac{7-9}{5} = \frac{3-9}{1} \quad \therefore 9 = 2S$$ （转移电导一定注意有单位!!!）

6. 相量图法：并联估构以并联模块电压 $\dot{U}$ 为基础相量.

支路1 纯电阻，$\dot{I}_1$ 与 $\dot{U}$ 同相.



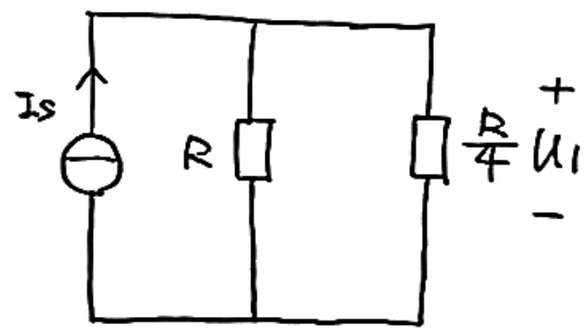
$\Longrightarrow$ $\omega \to 2\omega$，I_1 不变，I_2 变一半，I_3 变2倍.



Ⓐ的读数即 $\dot{I}$ 的模长，$I_A = \sqrt{5^2+40^2} = 5\sqrt{65}$ A.

7. $U = 5\sqrt{2}$，$\varphi = -135°$ $\therefore$ $u = 10\cos(10t - 135°)$ (V)

8. 把副边等效至原边：



$\Rightarrow u_1 = \frac{I_S R}{5}$

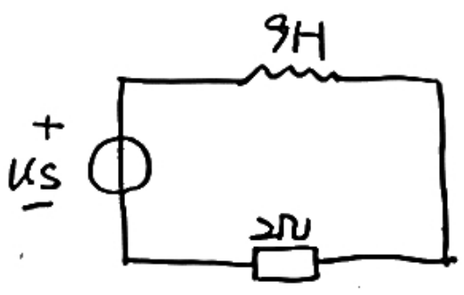
$\Rightarrow u_2 = 2u_1 = \frac{2I_S R}{5}$.

9. 列方程，略.（注意单位）.

10. 易得 $R = 8+4+2 = 14\Omega$.（把有伴电流源等效为有伴电压源即可）

$\therefore \tau = \frac{L}{R} = \frac{1}{14}$ s.

11. 先去耦：顺向串联. $L = L_1 + L_2 + 2M = 9$H.



三要素法：$i(t) = i(\infty) + [i(0^+) - i(\infty)]e^{-\frac{t}{\tau}}$.

$i(0^+) = i(0^-) = 0$.

$i(\infty) = \frac{1}{2}$

$\tau = \frac{L}{R} = \frac{9}{2}$ $\therefore$ $i(t) = \frac{1}{2} - \frac{1}{2}e^{-\frac{2}{9}t}$

$\therefore i(1) = \frac{1}{2} - \frac{1}{2}e^{-\frac{2}{9}}$ (A).

12. 三要素法 + 齐性定理：

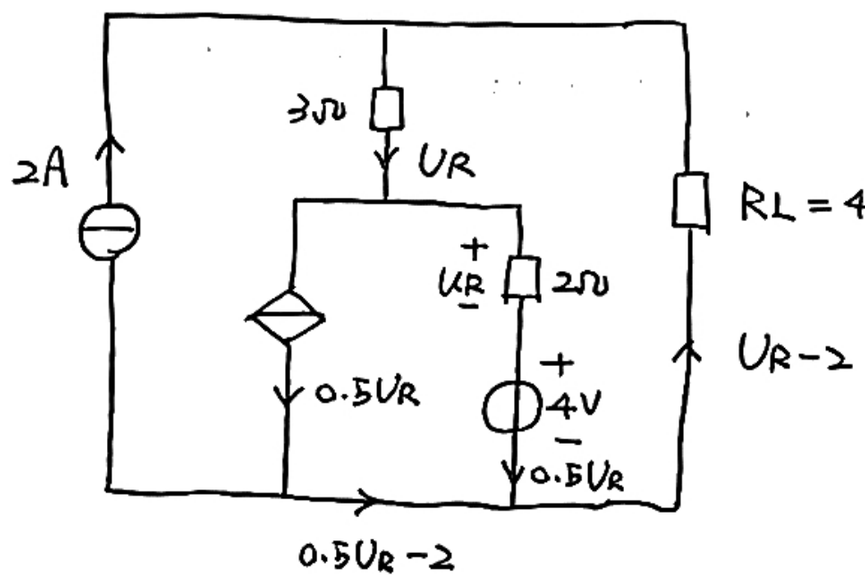
$$u(t) = u(\infty) + [u(0) - u(\infty)]e^{-t}.$$

Case1：$u_S(t) = 4\varepsilon(t)\,V$ 时，对比系数可知 $u(\infty) = 2V$，$u(0) = 0V$.

Case2：$u_S(t) = 8\varepsilon(t)\,V$ 时，由齐性定理：$u(\infty) = 2\times 2V = 4V$，条件中 $u_C(0) = 3V$

$$\therefore\ u_C(t) = 4 + (3-4)e^{-t} = 4 - e^{-t}\ (V).$$

二、 标电流 + 找回路列 KVL：



右侧回路KVL：

$$3U_R + U_R + 4 + 4(U_R - 2) = 0$$

$$\therefore\ U_R = \frac{1}{2}$$

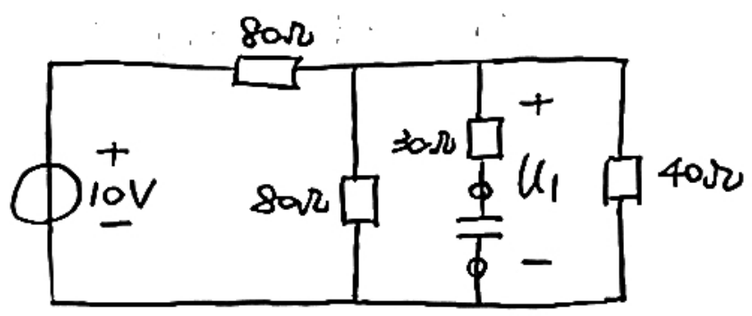
$$\therefore\ I_{RL} = \frac{3}{2}$$

$$P_{RL} = \left(\frac{3}{2}\right)^2 \cdot 4 = 9W.$$

三. 换路定理 + 戴维南.

$t=0^-$ 时，3Ω 电阻失效，$u_C(0^-) = 8\Omega$ 电阻电压 $= 5V = u_C(0^+)$.

闭合后，副边折至原边：

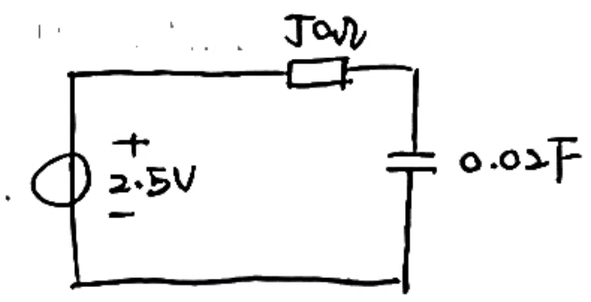


u_{OC}：3Ω 电阻失效. $R_1 = 80 // 40 = \frac{80}{3}\Omega$.

$$\therefore\ u_{OC} = \frac{10}{4} = 2.5V.$$

R_{eq}：10V 电源变导线：$R_{eq} = 80//80//40 + 30 = 50\Omega$.

$\Rightarrow$



$u_C(0^+) = 5V.$

$u_C(\infty) = 2.5V.$

$\tau = RC = 1S.$

$$\therefore\ u_C = 2.5 + 2.5e^{-t}\ (V).$$

$$i_C = C\cdot\frac{du_C}{dt} = -0.05e^{-t}.$$

$$\therefore\ u_1 = 30i_C + u_C = 2.5 + e^{-t}\ (V) \quad \therefore\ u = \frac{1}{2}u_1 = 1.25 + 0.5e^{-t}\ (V).$$

四. 1): $U = \sqrt{12^2 + 15^2 + 16^2} = 25V$.

2): ① 直流: $U_R = 12V$.

② 基波: L_1与C谐振, 并联谐振视作断路, $U_R = 0$

③ 二次谐波: $j2000 \,//\, -j500 + j\frac{2000}{3} = 0$.

($L_1 // C$ 与 L_2 串联谐振, 视作导线)

$\therefore \dot{U}_R = \dot{U}_S = 16\angle 0°$

$\therefore u_R(t) = 12 + 16\sqrt{2}\cos(2\omega t)$ (V).

3): 直流: $P_1 = \frac{12^2}{1} = 144W$.

基波: $P_2 = \frac{0^2}{1} = 0$

二次: $P_3 = \frac{16^2}{1} = 256W$

$P = P_1 + P_2 + P_3 = 400W$.

(平均功率只由R产生).

五. 级联. $T = T_1 T_2 = \begin{bmatrix} 1.5 & 10 \\ 0.05 & 1 \end{bmatrix}$, T参数使用代数法.

$$\begin{cases} u_1 = 100 - 80I_1. \\ u_2 = -20I_2 = u_R \\ u_1 = 1.5u_2 - 10I_2. \\ I_1 = 0.05u_2 - I_2 \end{cases} \Rightarrow \begin{cases} I_2 = -0.5. \\ u_R = 10V. \end{cases}$$

(Z、Y参数可考虑还原电路, A(T)与H参数一般代数法).